



MERU UNIVERSITY OF SCIENCE AND TECHNOLOGY

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UNIVERSITY EXAMINATIONS 2019/2020

FIRST YEAR SECOND SEMESTER SUPPLEMENTARY/SPECIAL EXAMINATION FOR THE CERTIFICATE IN INFORMATION TECHNOLOGY

CIT 1202: STRUCTURED PROGRAMMING

DATE: JANUARY 2021

TIME: 1½ HOURS

INSTRUCTIONS: Answer question **one** and any other **two** questions

QUESTION ONE (30 MARKS)

- a) Identify any three examples of computer hardware (3 marks)
- b) Programs store data temporarily in main memory. Define the term memory cell (2 marks)
- c) In memory a computer can either store a value or retrieve it. Distinguish between storing and retrieving values in computer memory (2 marks)
- d) Identify and discuss the two types of computer memory (4 marks)
- e) Secondary storage play an important role in computer systems. Identify two reasons why secondary storage is required (2 marks)
- f) Differentiate between computer hardware and software (2 marks)
- g) Identify any two roles played by the central processing unit (2 marks)
- h) Identify any four characteristics of a good algorithm (2 marks)
- i) C is a structured programming language. Explain why (2 marks)
- j) Discuss the role played by a C compiler (2 marks)
- k) A compiler is a translator. Distinguish between a compiler and an interpreter (4 marks)
- l) Part of the syntax of a C program is the pre-processor directives. Identify the role the pre-processor directives play in C (1 mark)
- m) The printf() function is defined in the standard input output library. What is the role of the printf() function (1 mark)

- n) The semi colon is important when declaring variables and statements. What role does a semi colon play in a variable declaration (1 mark)

QUESTION TWO (15 MARKS)

- a) C language contains reserved words. Briefly describe your understanding of reserved words (2 marks)
- b) Identify any 5 reserved words in C (5 marks)
- c) Describe the two types of identifiers in C (4 marks)
- d) Identify any four identifier naming rules in C (4 marks)

QUESTION THREE (15 MARKS)

- a) You are an IT technician in a health clinic. You have been asked to write a sample C program that would help the doctor capture some data about a patient. Ensure your program captures data pertaining to: patient's name, age, eye colour, weight , skin colour and hair colour (6 marks)
- b) Write a program in C that stores marks for 5 students in an array. The program should contain a function that calculates the sum and average of the marks in the array (6 marks)
- c) The following is a C variable declaration statement. Write a pointer to the variable int studentMark=30 (1 mark)
- d) Identify any two data types in C (2 marks)

QUESTION FOUR (15 MARKS)

- a) Discuss any three common programming errors (6 marks)
- b) Assuming you are an examination officer at your school, write a C program that receives a number between 0 and 100 and displays on the screen the grade for that number (6 marks)
- c) Identify the role of the scanf() function in C
- d) At the weighing office in a coffee factory, farmers bring their produce in bags. Having been newly hired as a programmer, identify the appropriate datatypes you

would use to represent each of the followed: bags of maize each farmer has brought;
total weight of coffee each farmer has brought (2 marks)

QUESTION FIVE (15 MARKS)

- a) Identify any two type of functions in C (4 marks)
- b) Write a brief C program that opens a file called”programming.txt” (4 marks)
- c) Briefly illustrate the structure of a for loop in C (2 marks)
- d) The table below represents litres of milk a farmer sells each day

Day	Litres
1	34
2	50
3	45
4	40

Being a programmer, write a program in C that calculates the average litres sold by the
farmer for the four days (4 marks)

- e) In your own understanding, illustrate the application of the **break** keyword (1 mark)